

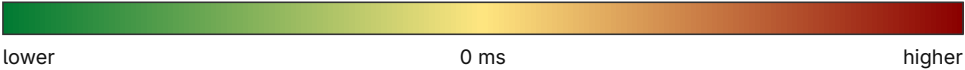
Query 01 - Insert Last Child under Last Root

Graph	Parameters	Baseline (ms)	Dewey (ms)	Prepost (ms)	Dewey cost (ms)	Prepost cost (ms)
F40	Last root, last child position	2.060	3.769	2.699	+1.709	+0.639
F1000	Last root, last child position	2.450	4.965	4.217	+2.515	+1.767
NT10	Last root, last child position	2.017	3.708	2.782	+1.691	+0.765
NT100	Last root, last child position	2.371	3.388	2.974	+1.017	+0.603
NT1000	Last root, last child position	2.775	4.196	4.210	+1.421	+1.435
NT10000	Last root, last child position	6.369	15.040	18.584	+8.671	+12.215
NT100000	Last root, last child position	34.737	95.469	157.723	+60.732	+122.986
DT10	Last root, last child position	2.086	3.123	2.708	+1.037	+0.622
DT100	Last root, last child position	2.196	3.656	2.925	+1.460	+0.729
DT1000	Last root, last child position	2.237	4.679	4.567	+2.442	+2.330
DT10000	Last root, last child position	6.190	15.620	17.949	+9.430	+11.759
DT100000	Last root, last child position	34.508	96.526	157.941	+62.018	+123.433
WT10	Last root, last child position	2.267	3.542	3.083	+1.275	+0.816
WT100	Last root, last child position	2.157	3.626	3.084	+1.469	+0.927
WT1000	Last root, last child position	2.311	4.870	4.551	+2.559	+2.240
WT10000	Last root, last child position	6.065	15.182	18.370	+9.117	+12.305
WT100000	Last root, last child position	33.532	93.953	159.105	+60.421	+125.573
SNB/C	Last root, last child position	615.404	2,379.970	2,828.895	+1,764.566	+2,213.491
SNB/P	Last root, last child position	2.695	5.550	5.012	+2.855	+2.317
SNB/T	Last root, last child position	2.045	3.241	2.726	+1.196	+0.681

Graph: F = forest; NT = normal tree (truebase); DT = deep tree (ultrataill); WT = wide tree (ultrawide); number = node count; SNB/C = Comment, SNB/P = Place, SNB/T = Tagclass (SNB SF1).

Runtimes: median of five runs. Maintenance cost = indexed representation - Baseline, in milliseconds (ms).

Cost colors: green = lower than Baseline; yellow = zero; red = higher than Baseline. Intensity uses a logarithmic global scale.



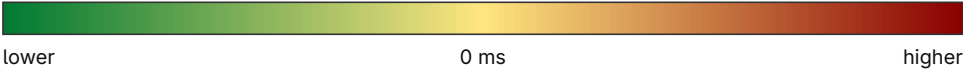
Query 02 - Insert First Child under First Root

Graph	Parameters	Baseline (ms)	Dewey (ms)	Prepost (ms)	Dewey cost (ms)	Prepost cost (ms)
F40	First root, first child position	2.109	3.660	3.049	+1.551	+0.940
F1000	First root, first child position	2.688	8.060	14.161	+5.372	+11.473
NT10	First root, first child position	1.976	3.073	2.798	+1.097	+0.822
NT100	First root, first child position	2.235	4.893	3.880	+2.658	+1.645
NT1000	First root, first child position	2.675	21.066	15.428	+18.391	+12.753
NT10000	First root, first child position	5.964	178.283	108.927	+172.319	+102.963
NT100000	First root, first child position	35.622	2,706.851	1,279.879	+2,671.229	+1,244.257
DT10	First root, first child position	1.938	3.402	2.969	+1.464	+1.031
DT100	First root, first child position	2.291	5.221	3.729	+2.930	+1.438
DT1000	First root, first child position	2.555	23.181	14.240	+20.626	+11.685
DT10000	First root, first child position	6.077	192.295	97.207	+186.218	+91.130
DT100000	First root, first child position	35.456	2,643.025	1,064.157	+2,607.569	+1,028.701
WT10	First root, first child position	2.083	3.850	3.108	+1.767	+1.025
WT100	First root, first child position	2.187	4.879	4.192	+2.692	+2.005
WT1000	First root, first child position	2.622	20.776	17.254	+18.154	+14.632
WT10000	First root, first child position	5.946	173.464	117.772	+167.518	+111.826
WT100000	First root, first child position	33.742	2,310.665	1,312.333	+2,276.923	+1,278.591
SNB/C	First root, first child position	614.627	2,701.642	29,898.790	+2,087.015	+29,284.163
SNB/P	First root, first child position	2.662	21.352	23.048	+18.690	+20.386
SNB/T	First root, first child position	1.989	4.252	3.301	+2.263	+1.312

Graph: F = forest; NT = normal tree (truebase); DT = deep tree (ultrataill); WT = wide tree (ultrawide); number = node count; SNB/C = Comment, SNB/P = Place, SNB/T = Tagclass (SNB SF1).

Runtimes: median of five runs. Maintenance cost = indexed representation - Baseline, in milliseconds (ms).

Cost colors: green = lower than Baseline; yellow = zero; red = higher than Baseline. Intensity uses a logarithmic global scale.



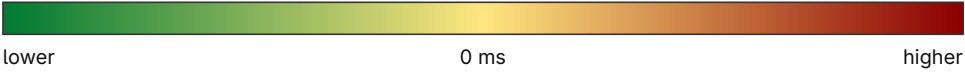
Query 03 - Insert Last Root

Graph	Parameters	Baseline (ms)	Dewey (ms)	Prepost (ms)	Dewey cost (ms)	Prepost cost (ms)
F40	Last root position	1.408	2.255	1.933	+0.847	+0.525
F1000	Last root position	1.519	2.015	1.945	+0.496	+0.426
NT10	Last root position	1.406	2.529	1.922	+1.123	+0.516
NT100	Last root position	1.627	2.020	1.964	+0.393	+0.337
NT1000	Last root position	1.591	2.279	2.052	+0.688	+0.461
NT10000	Last root position	1.504	2.324	2.036	+0.820	+0.532
NT100000	Last root position	1.582	2.242	1.365	+0.660	-0.217
DT10	Last root position	1.296	2.054	1.807	+0.758	+0.511
DT100	Last root position	1.570	2.214	1.913	+0.644	+0.343
DT1000	Last root position	1.494	2.351	2.000	+0.857	+0.506
DT10000	Last root position	1.470	2.265	2.193	+0.795	+0.723
DT100000	Last root position	1.628	2.252	1.938	+0.624	+0.310
WT10	Last root position	1.477	1.697	2.272	+0.220	+0.795
WT100	Last root position	1.584	2.359	2.143	+0.775	+0.559
WT1000	Last root position	1.608	2.407	2.271	+0.799	+0.663
WT10000	Last root position	1.512	1.989	1.931	+0.477	+0.419
WT100000	Last root position	1.458	2.128	2.203	+0.670	+0.745
SNB/C	Last root position	1.610	168.056	2.013	+166.446	+0.403
SNB/P	Last root position	1.285	1.974	1.791	+0.689	+0.506
SNB/T	Last root position	1.295	2.054	1.854	+0.759	+0.559

Graph: F = forest; NT = normal tree (truebase); DT = deep tree (ultrataill); WT = wide tree (ultrawide); number = node count; SNB/C = Comment, SNB/P = Place, SNB/T = Tagclass (SNB SF1).

Runtimes: median of five runs. Maintenance cost = indexed representation - Baseline, in milliseconds (ms).

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Query 04 - Insert First Root

Graph	Parameters	Baseline (ms)	Dewey (ms)	Prepost (ms)	Dewey cost (ms)	Prepost cost (ms)
F40	First root position	1.462	2.891	2.183	+1.429	+0.721
F1000	First root position	1.550	18.873	12.676	+17.323	+11.126
NT10	First root position	1.483	2.695	1.910	+1.212	+0.427
NT100	First root position	1.530	3.772	2.881	+2.242	+1.351
NT1000	First root position	1.616	19.092	14.358	+17.476	+12.742
NT10000	First root position	1.560	163.056	100.699	+161.496	+99.139
NT100000	First root position	1.450	2,468.514	1,190.367	+2,467.064	+1,188.917
DT10	First root position	1.472	2.704	1.965	+1.232	+0.493
DT100	First root position	1.534	4.277	3.032	+2.743	+1.498
DT1000	First root position	1.506	21.538	13.091	+20.032	+11.585
DT10000	First root position	1.616	175.722	88.245	+174.106	+86.629
DT100000	First root position	1.584	2,408.674	951.062	+2,407.090	+949.478
WT10	First root position	1.541	2.720	2.111	+1.179	+0.570
WT100	First root position	1.332	3.910	3.228	+2.578	+1.896
WT1000	First root position	1.629	18.067	15.416	+16.438	+13.787
WT10000	First root position	1.121	157.326	107.311	+156.205	+106.190
WT100000	First root position	1.654	2,097.428	1,195.213	+2,095.774	+1,193.559
SNB/C	First root position	1.442	47,233.966	28,444.850	+47,232.524	+28,443.408
SNB/P	First root position	1.315	25.599	21.048	+24.284	+19.733
SNB/T	First root position	1.242	3.311	2.464	+2.069	+1.222

Graph: F = forest; NT = normal tree (truebase); DT = deep tree (ultrataill); WT = wide tree (ultrawide); number = node count; SNB/C = Comment, SNB/P = Place, SNB/T = Tagclass (SNB SF1).

Runtimes: median of five runs. Maintenance cost = indexed representation - Baseline, in milliseconds (ms).

Cost colors: green = lower than Baseline; yellow = zero; red = higher than Baseline. Intensity uses a logarithmic global scale.

